Project Report

1. PROJECT TITLE

VidyaMitra: The Intelligent Career Agent – AI-Powered Resume Evaluator, Trainer & Career Planner

VidyaMitra is an AI-based career assistance platform designed to help students and job seekers improve their employability. The system analyzes resumes, identifies skill gaps, and provides personalized recommendations for skill development, interview preparation, and career planning.
It acts as a virtual career mentor by offering resume evaluation, learning suggestions, mock interview practice, and career progress tracking. The goal is to bridge the gap between academic learning and industry requirements.

2. PROJECT DESCRIPTION

● System Overview

Vidyamitra is an AI-powered career guidance system that helps users improve their resumes, identify skill gaps, practice interviews, and plan career growth. The system analyzes resumes using AI models and provides suggestions for improvement, training recommendations, and interview preparation.

● Problem Domain

Many students and job seekers struggle with:

- Poor resume formatting
- Lack of ATS-optimized keywords
- Missing industry-required skills
- Limited interview preparation

Vidyamitra solves these problems by providing automated AI-based career guidance.

● Core Technology Used

Frontend Technologies:

- HTML

- Tailwind CSS
- Vite
- JavaScript

Backend Technologies:

- Python
- FastAPI / API-based backend architecture

Database / Cloud:

- Supabase

Other Technologies:

- AI model APIs
- REST APIs
- Environment configuration using .env

● Key Capabilities

- AI-based resume analysis
- Skill gap detection
- ATS keyword optimization
- Interview question generation
- Career path suggestions
- Training and learning recommendations

● Target Users

- College students preparing for placements
- Final-year students entering the job market
- Fresh graduates
- Job seekers improving resumes and interview skills
- Training institutes

● Real-world Relevance

Modern hiring systems rely heavily on automated resume screening tools. Vidyamitra helps candidates optimize their resumes and develop relevant skills to improve employability and interview success.

3. APPLICATION SCENARIOS / USE CASES

Educational Usage

Students preparing for campus placements can analyze their resumes and receive suggestions to improve job readiness.

End-user Usage

Job seekers can upload resumes, identify missing skills, and receive interview preparation guidance.

Enterprise Usage

HR teams can integrate the system to automatically analyze candidate resumes and shortlist suitable applicants.

Industrial Usage

Training institutes can use the platform to guide students in developing industry-relevant skills.

4. TECHNICAL ARCHITECTURE OVERVIEW

● High-level Architecture Diagram

User

|

Frontend (HTML + Tailwind + Vite)

|

Backend API (Python)

|

AI Resume Analysis Engine

|

Supabase Database

|

Results Returned to Frontend

● Component Interaction Diagram

Main components include:

User Interface

|

Frontend Application

|

Backend API (main.py)

|

┌──────────────────────┐

│ AI Model APIs |

└──────────────────────┘

|

Supabase Database

|

Response Sent to User

● Data / Request Flow

1. User opens the Vidyamitra web interface.
2. User uploads a resume file.
3. Frontend sends the resume to the backend API.
4. Backend processes the request.
5. AI model analyzes the resume.
6. Skill gaps and recommendations are generated.
7. Results are returned to the frontend.
8. User receives feedback and improvement suggestions.
9. Analysis displayed on dashboard

System Layers Explanation

Frontend Layer

Responsible for user interaction and displaying results.

Key files:

- index.html
- src/
- tailwind.config.js
- vite.config.js

Features:

- Resume upload interface
- Result display dashboard
- User interaction interface

Backend Layer

Handles system logic and AI processing.

Important files:

- main.py
- check_supabase.py
- diagnose_keys.py
- list_models.py

Responsibilities:

- API request processing
- Resume analysis
- AI model integration
- Database interaction

Database Layer

Supabase cloud database stores:

- User data
- Resume analysis records
- Skill recommendations

External APIs

- AI model APIs
- Supabase cloud database

Deployment Environment

- Web-based system
- Node.js frontend server
- Python backend server

5. PREREQUISITES

5.1 Software Requirements

Development Tools

- Python 3.x
- Node.js
- npm

IDE

- Visual Studio Code

Runtime Environment

- Python interpreter
- Node.js runtime

Cloud Services

- Supabase

5.2 Libraries / Frameworks / Dependencies

Backend:

- fastapi

- python-dotenv
- supabase
- requests

Frontend:

- vite
- tailwindcss
- postcss

Other Dependencies:

- Node modules
- Environment configuration tools

5.3 Hardware Requirements

Minimum system requirements:

- 8GB RAM
- Intel i5 processor or higher
- Internet connection
- 10GB storage

6. PRIOR KNOWLEDGE REQUIRED

Programming Languages

- Python
- JavaScript
- HTML and CSS

Framework Basics

- Web development fundamentals
- REST APIs

Database Fundamentals

- Basic database concepts
- Cloud databases

Networking / Cloud Concepts

- API communication

AI / ML Fundamentals

- Natural Language Processing basics
- AI model integration

7. PROJECT OBJECTIVES

Technical Objectives

Develop an AI-powered system capable of analyzing resumes and recommending improvements.

Performance Objectives

Ensure fast processing and accurate suggestions.

Deployment Objectives

Create a scalable web-based career guidance platform.

Learning Outcomes

- Full-stack web development
- AI integration in applications
- API development
- Cloud database integration

8. SYSTEM WORKFLOW

Step-by-step execution:

1. User opens the Vidyamitra platform.
2. User uploads a resume file.
3. Frontend sends the file to backend API.
4. Backend processes resume data.
5. AI engine analyzes content.
6. Skill gaps and suggestions are generated.
7. Results are displayed on the frontend.

9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

9.1 Problem Definition

Job seekers face difficulties creating optimized resumes and identifying required skills for job roles.

9.2 Functional Requirements

The system should:

- Accept resume uploads
- Analyze resume content
- Detect skill gaps
- Provide interview questions
- Suggest learning resources

9.3 Non-Functional Requirements

- Fast response time
- Secure data storage
- Scalable architecture
- High system availability

9.4 System Design Decisions

- Web-based application
- API-driven backend

- Cloud database
- AI-powered resume analysis

9.5 Technology Stack Selection

Layer	Technology
Frontend	HTML, Tailwind, Vite
Backend	Python
Database	Supabase
AI	AI model APIs

10. MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION

10.1 Development Environment Setup

Install:

- Python
- Node.js
- VS Code

10.2 Dependency Installation

Backend dependencies

pip install -r requirements.txt

Frontend dependencies

npm install

10.3 Project Structure Creation

Project Folder Structure

```
vidyamitra/
│
├── backend/
│
├── frontend/
│
├── .env
│
└── README.md
```

10.4 Configuration Setup

The configuration of the VidyaMitra system is managed using the .env file, which stores important environment variables such as API keys, database connections, and other system settings. This helps keep sensitive information secure and separate from the main code.

Configuration Files:

- backend/ – Contains server-side logic and AI processing modules.
- frontend/ – Contains the user interface of the application.
- .env – Stores environment variables and configuration settings.
- README.md – Provides project documentation and setup instructions.

11. MILESTONE 3: CORE SYSTEM DEVELOPMENT

Feature 1 – Resume Analyzer

Description
Analyzes resumes to identify missing skills and formatting issues.

Code Logic

- Upload resume
- Extract text

- Analyze keywords
- Generate feedback

Output

- Resume score
- Improvement suggestions

Feature 2 – Skill Gap Detection

The system compares resume skills with industry requirements and recommends missing skills.

Feature 3 – Mock Interview Generator

The system generates technical and HR interview questions based on user skills.

API Endpoints

Example endpoints:

POST /upload_resume
GET /analyze_resume
GET /skill_suggestions

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

Component Integration

Frontend connected with backend APIs.

Performance Optimization

- Faster API processing
- Efficient data handling

Security Enhancements

- Environment variables for API keys
- Secure API requests

Error Handling

- Invalid resume format detection
- Server error management

13. MILESTONE 5: TESTING & VALIDATION

Test Cases

Test Case ID	Input	Expected Output	Status
TC1	Upload resume	Resume analyzed	Pass
TC2	Missing skills	Skill suggestions	Pass
TC3	Interview generator	Questions generated	Pass

Unit Testing

Individual backend functions tested.

Integration Testing

Frontend and backend communication verified.

User Acceptance Testing

Students tested system usability and feedback quality.

Performance Metrics

Response time:
2–4 seconds per analysis

Load Handling:
Supports multiple users simultaneously

Security Validation:
API keys secured using environment variables

14. DEPLOYMENT

Deployment Architecture

```
User Browser
      |
Frontend Server
      |
Backend Python API
      |
AI Model Service
      |
Supabase Database
```

Hosting Platform

Possible platforms:

- Vercel (frontend)
- Render / Railway (backend)

Deployment Steps

1. Deploy frontend build
2. Deploy backend API
3. Configure environment variables
4. Connect Supabase database

Production Considerations

- Secure API keys
- Scalable server infrastructure
- Database backup strategy

15. PROJECT STRUCTURE

```
project_root
│
├── backend
│   ├── __pycache__
│   ├── data
│   ├── middleware
│   ├── routes
│   ├── services
│   ├── .env
│   ├── check_supabase.py
│   ├── diagnose_keys.py
│   ├── list_models.py
│   ├── main.py
│   └── requirements.txt
│
├── frontend
│   ├── node_modules
│   ├── public
│   ├── src
│   ├── .env
│   ├── index.html
│   ├── package.json
```

```
│   ├── postcss.config.js
│   ├── tailwind.config.js
│   └── vite.config.js
│
├── AI_SETUP_GUIDE.md
├── README.md
├── start-backend.bat
└── TODO.md
```

Folder Explanation

backend
Contains backend API logic and AI integration.

frontend
Contains UI code and web interface.

routes
Handles API endpoints.

services
Contains business logic and AI functions.

middleware
Handles request processing and authentication.

data
Stores processing data or models.

16. RESULTS

System Output:

- Resume evaluation report

- Skill recommendations
- Interview practice questions

Performance:

- Fast response time
- Accurate suggestions

Screenshots:

- Resume upload interface
- Analysis result dashboard

17. ADVANTAGES & LIMITATIONS

Advantages

- Automated resume analysis
- Personalized career guidance
- AI-powered recommendations
- Time-saving for job seekers

Limitations

- AI accuracy depends on training data
- Requires internet connection
- Limited job role dataset

18. FUTURE ENHANCEMENTS

- Mobile application development
- Advanced AI career prediction
- LinkedIn integration
- Job portal integration
- Automated resume builder
- Real-time interview simulator

19. CONCLUSION

The Vidyamitra system demonstrates how artificial intelligence can enhance career guidance by analyzing resumes and identifying skill gaps. The platform

integrates frontend, backend, AI models, and cloud databases to provide intelligent recommendations for job seekers. This project highlights the practical application of AI and full-stack development in solving real-world career preparation challenges.

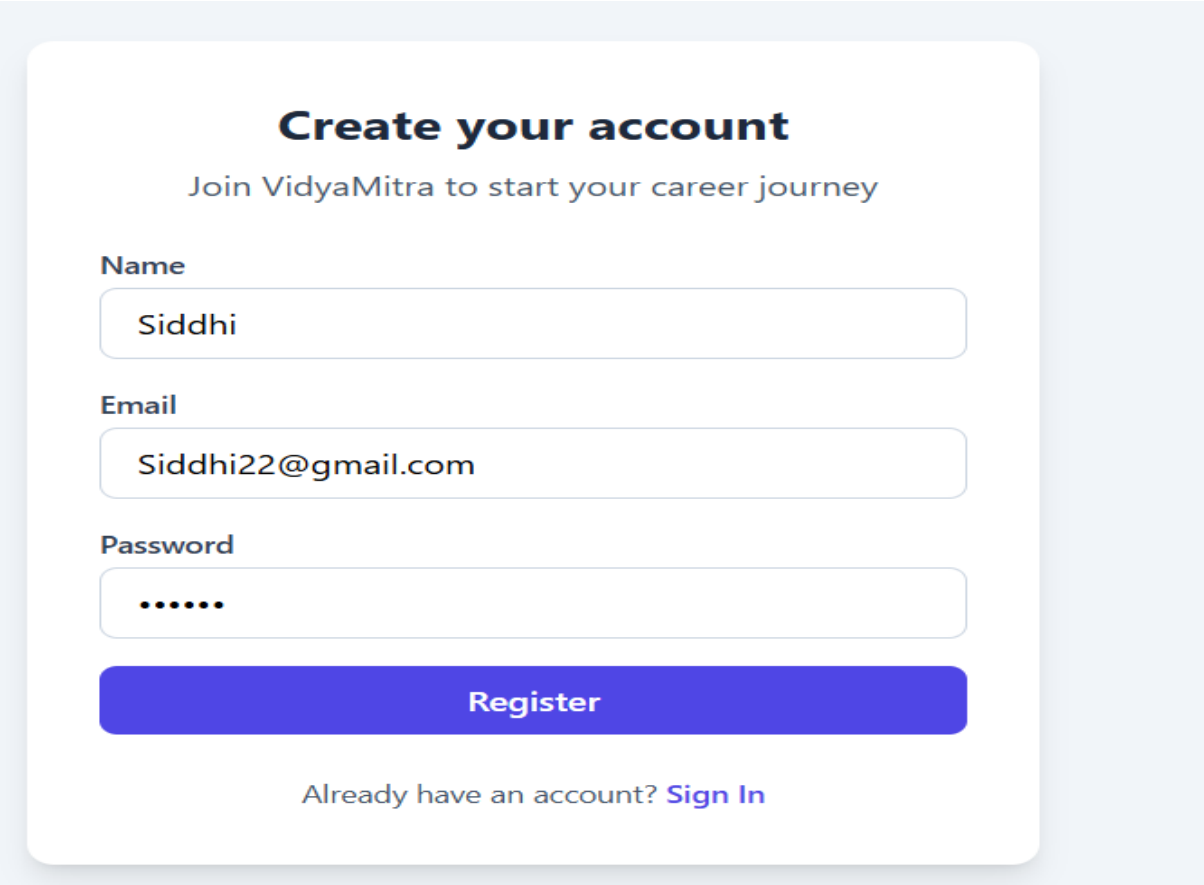
20. APPENDIX

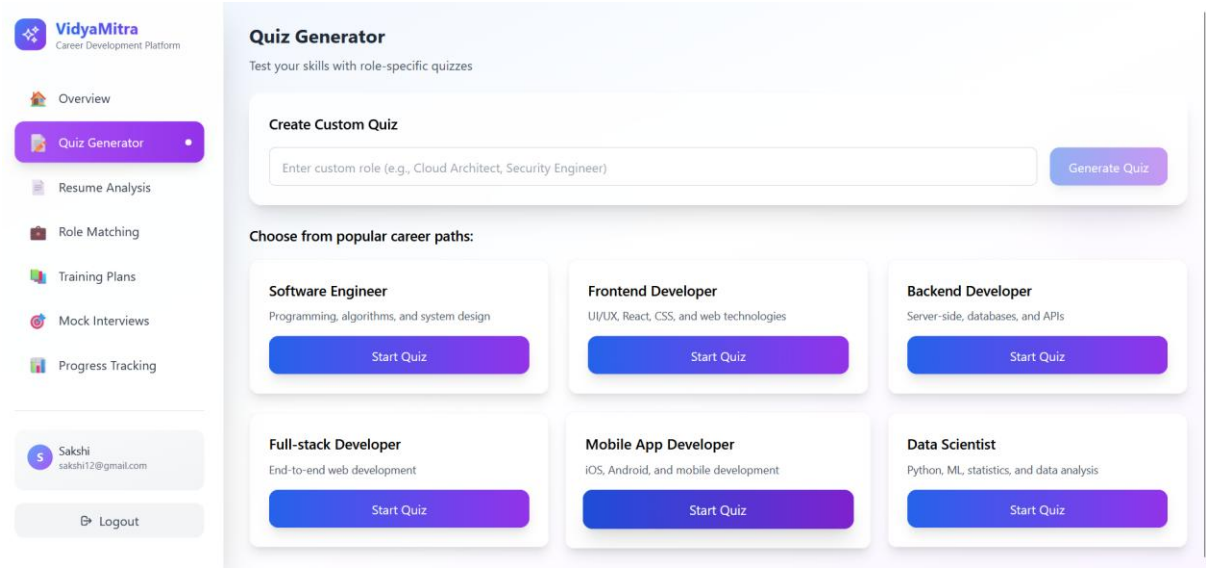
Source Code/Github Repo:

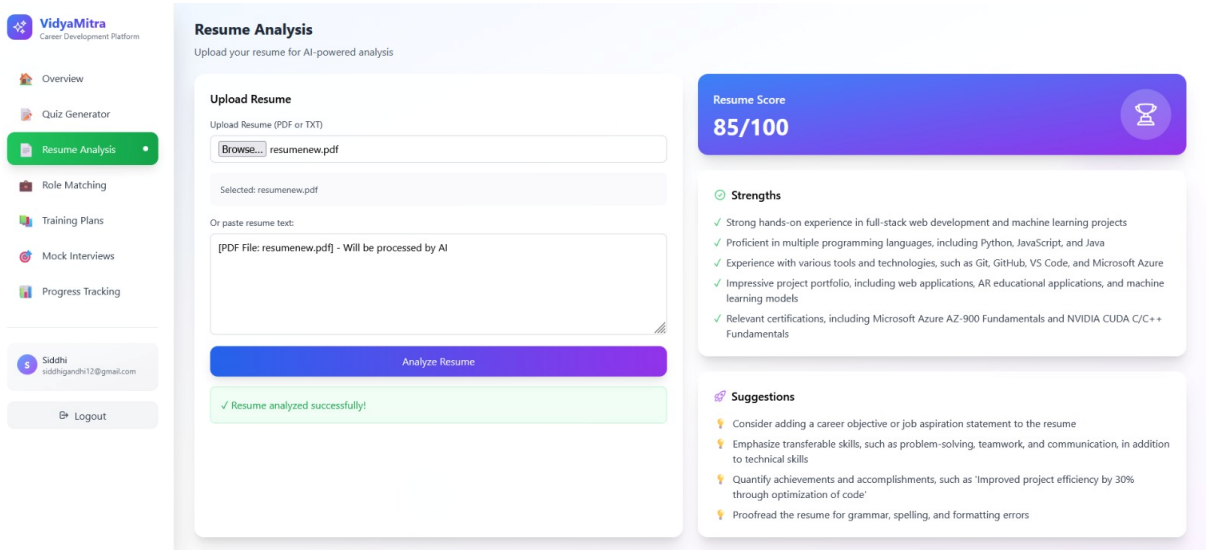
https://github.com/gandhisiddhi81/Vidya_Mitra_Career_Devlopment_Platform-Frontend-

https://github.com/gandhisiddhi81/Vidya_Mitra_Career_Devlopment_Platform-Backend-

Output Images




VidyaMitra
Career Development Platform
Overview
Quiz Generator
Resume Analysis
Role Matching
Training Plans
Mock Interviews
Progress Tracking
Sakshi
sakshi12@gmail.com
Logout
Quiz Generator
Test your skills with role-specific quizzes
Create Custom Quiz
Enter custom role (e.g., Cloud Architect, Security Engineer)
Generate Quiz
Choose from popular career paths:
Software Engineer
Programming, algorithms, and system design
Start Quiz
Frontend Developer
UI/UX, React, CSS, and web technologies
Start Quiz
Backend Developer
Server-side, databases, and APIs
Start Quiz
Full-stack Developer
End-to-end web development
Start Quiz
Mobile App Developer
iOS, Android, and mobile development
Start Quiz
Data Scientist
Python, ML, statistics, and data analysis
Start Quiz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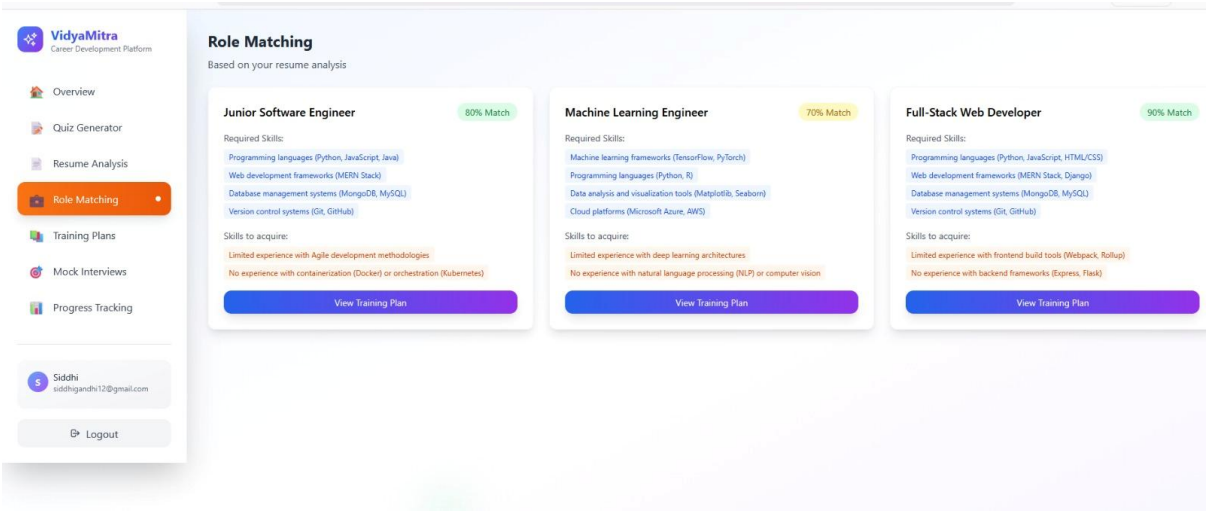
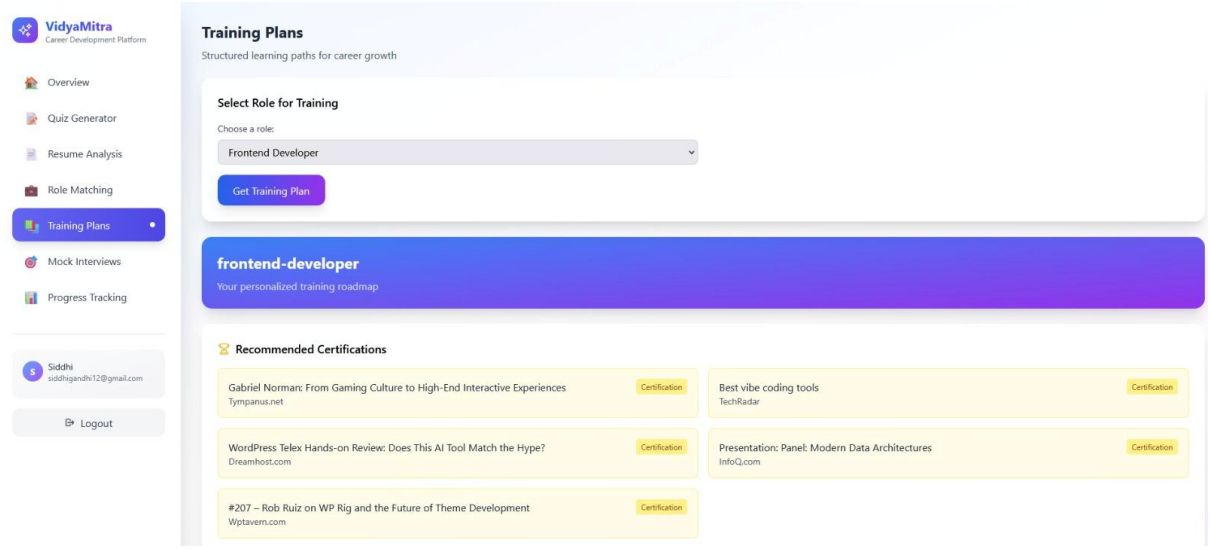
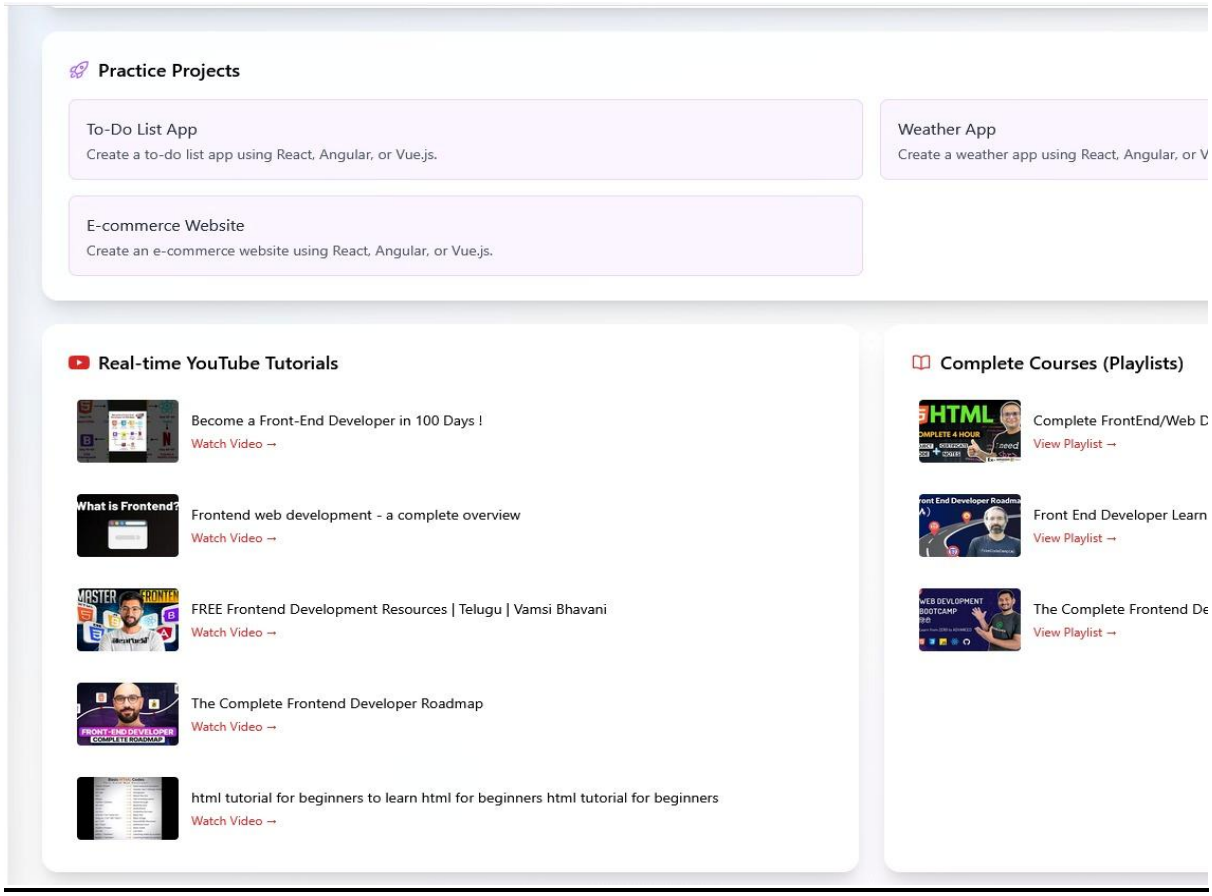
siddhigandhi12@gmail.com
Logout
Resume Analysis
Upload your resume for AI-powered analysis
Upload Resume
Upload Resume (PDF or TXT)
Browse... resumenew.pdf
Selected: resumenew.pdf
Or paste resume text:
[PDF File: resumenew.pdf] - Will be processed by AI
Analyze Resume
✓ Resume analyzed successfully!
Resume Score
85/100
Strengths
✓ Strong hands-on experience in full-stack web development and machine learning projects
✓ Proficient in multiple programming languages, including Python, JavaScript, and Java
✓ Experience with various tools and technologies, such as Git, GitHub, VS Code, and Microsoft Azure
✓ Impressive project portfolio, including web applications, AR educational applications, and machine learning models
✓ Relevant certifications, including Microsoft Azure AZ-900 Fundamentals and NVIDIA CUDA C/C++ Fundamentals
Suggestions
Consider adding a career objective or job aspiration statement to the resume
Emphasize transferable skills, such as problem-solving, teamwork, and communication, in addition to technical skills
Quantify achievements and accomplishments, such as 'Improved project efficiency by 30% through optimization of code'
Proofread the resume for grammar, spelling, and formatting errors


VidyaMitra
Career Development Platform
Overview
Quiz Generator
Resume Analysis
Role Matching
Training Plans
Mock Interviews
Progress Tracking
Siddhi
siddhigandhi12@gmail.com
Logout
Role Matching
Based on your resume analysis
Junior Software Engineer
80% Match
Required Skills:
Programming languages (Python, JavaScript, Java)
Web development frameworks (MERN Stack)
Database management systems (MongoDB, MySQL)
Version control systems (Git, GitHub)
Skills to acquire:
Limited experience with Agile development methodologies
No experience with containerization (Docker) or orchestration (Kubernetes)
View Training Plan
Machine Learning Engineer
70% Match
Required Skills:
Machine learning frameworks (TensorFlow, PyTorch)
Programming languages (Python, R)
Data analysis and visualization tools (Matplotlib, Seaborn)
Cloud platforms (Microsoft Azure, AWS)
Skills to acquire:
Limited experience with deep learning architectures
No experience with natural language processing (NLP) or computer vision
View Training Plan
Full-Stack Web Developer
90% Match
Required Skills:
Programming languages (Python, JavaScript, HTML/CSS)
Web development frameworks (MERN Stack, Django)
Database management systems (MongoDB, MySQL)
Version control systems (Git, GitHub)
Skills to acquire:
Limited experience with frontend build tools (Webpack, Rollup)
No experience with backend frameworks (Express, Flask)
View Training Plan





Video Link: https://drive.google.com/file/d/1OEHVHIaxQRvlejxBMEep-NoVp2EP9mwj/view?usp=drive_link

Render Deployment Link:

https://vidya-mitra-career-devlopment-platform-r457.onrender.com